Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/15/2006
PI Name:	Osheroff, Douglas Dean Ph.D.		
Project Title:	Tests of Fundamental Physics Through Studies of Superfluid Helium Three		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-4060	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	There were two personnel on this project, Victor Tsepelin, a postdoc, and Arito Nishimori, a grad student. Tsepelin left in the spring of 2004 and Nishimori, having gotten his Ph.D., left in February 2006. I am all that is left.		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2888		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
	This task will utilize the unique properties of superfluid helium three in a ground based study to test two specific predictions of fundamental physical theories. The first is a test of the theory which is used to explain and predict the properties thermodynamic systems undergoing second order phase transitions. The second is the theory of the parity-violating weak interaction and the specific prediction that one consequence of this interaction is the existence of a permanent electric dipole moment in certain particles or systems, thereby breaking the symmetry of these entities. The nature of this program is two fold. The first is to investigate the nature of superfluid 3He within the narrow region near the superfluid transition temperature, Tc, in which fluctuations in the properties of the superfluid diverge, and should dominate the properties and even the symmetry of the ordered phase. For most second order phase transitions these critical fluctuations determine the evolution of the system from a disordered to an ordered state by determining the		

Task Description:	temperature dependence of the various properties of the ordered state. In the case of superfluid 3He, however, the ordered state is much more complex, and the fluctuations are predicted to stabilize a specific state near Tc that is not stable at lower temperatures, and indeed has never been seen before. Identifying the nature of the superfluid within the critical regime would thus be a unique test of the renormalization group theory that was developed by Ken Wilson to understand and explain the behavior in the critical regime, and for which he was awarded the Nobel Prize in 1982. The critical regime in superfluid 3He is predicted to be very narrow, although a number of experimental results suggest that the effects of fluctuations are much stronger than one might expect, and occur over a much wider temperature range than predicted. Thus the first test of this project is to investigate the width of the critical regime, using careful measurements of thermodynamic properties of a sample in which the thermal gradients across the sample are much less than a billionth of a degree. The second and much more ambitious goal of the project will be to determine the nature of the superfluid state within the critical regime. To do so, we will employ a combination of nuclear magnetic resonance (NMR) studies, which are sensitive to the symmetry of the ordered state, and static magnetization measurements, which can help differentiate between certain specific states. If these two goals are met, there will be strong motivation to continue such studies in microgravity, where the smearing of the transition temperature due to the varying hydrostatic pressure throughout the sample can be eliminated. The second portion of this program will test a very different physical theory, that being a prediction that the weak force in nature should create permanent electric dipole moments in many different particles, thus breaking the symmetry that they would otherwise have. This produces a violation of parity, according to the theory first developed by Lee and Yang, and for which these two shared the Nobel Prize in 1957. Despite the very long time since the development of this theory and the specific prediction of the formation of such permanent electric dipole moments, no such moments have ever been detected, owing to their extremely small magnitude. One expects the atoms in a sample of superfluid 3He to possess such a week electric dipole moment, but here the ordered superfluid state should oriented all the tiny electric dipole moments in the sample along a specific axis in the low temperature superfluid phase (the B phase). This axis, which is only weakly, affected externally applied fields (electric, magnetic, flow, or thermal gradient) across the sample. The NMR frequency in the B liquid, however, is sensitive to the relative orientations of this axis and the static applied magnetic field. In our experiment, we will not be sensitive to the orientation of this axis over the entire sample, but predominantly that portion of the sample furthest from the electrodes that we must use to apply a large electric field across the sample. If the there is a permanent electric dipole moment along this axis, the application of the electric field should alter its orientation with respect to the static NMR field, and thus cause a shift in the NMR frequency of the liquid. The shift in NMR frequency due to a permanent electric dipole moment will be proportional to the magnitude and sign of the electric field. While this second experiment does not require microgravity, it will add greatly to the value of this project if it succeeds. If successful, this task will substantially strengthen the degree to which these two important fundamental physical theories (that of critical phenomena and that of the parity violation) have been tested and confirmed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If this project had been successful this work would have provided a better test on certain theories of phase transitions that might be important in chemical industry or perhaps in certain implementations of quantum computing.
Task Progress:	The postdoc hired to work on this NASA grant, Victor Tsepelin, left after his three years were up, in the late spring of 2004. After that time my grad student, Arito Nisimori, was the only person working on the project (besides myself); however, progress has been good. At the time the postdoc left we had developed enough resolution with our SQUID magnetometer to measure the jump in magnetization of liquid 3He at Tc reported earlier by the Wheatley group, and it appeared that we were seeing about the same size jump (about 0.5%). However, once the project was in the hands Arito, I asked him to run the experiment differently. In the Wheatley experiment and in our earlier experiments, the cell was allowed to drift warmer slowly, so the walls of the cell would be roughly in thermal equilibrium with the liquid 3He. In our cell, the walls were made of bronze, and heat sunk separately to the copper nuclear cooling stage. I suggested that Arito keep the cell walls at the starting temperature, below Tc, and that a heat load be generated in a heater immersed within the liquid in order to warm it through Tc, while the cell itself remain several microkelvins below Tc. Once this was done it became clear that most of the change in magnetization seen near Tc was associated with magnetization in the walls of both sample chambers. If one calculates the expected jump at Tc, it should only be about 0.1%, which as only a couple of times our noise limit due to instabilities. In a conversation with Bruce Patton, whose calculation has stimulated our proposal in the first place, it was pointed out that the same physics would suggest there should be a long high temperature tail due to critical fluctuations on the NMR frequency shift in the A-liquid above Tc, directly proportional to the tail seen in the magnetization jump. This has a great advantage in that the entire signal comes from the liquid 3He, and hence cannot be confused with a background signal. Indeed, no other nuclear spin system exhibits a frequency shift at constant magnetic field. To execute this experiment it was necessary to totally change our cooling technique, as the fringing field from the nuclear demag stage would cause its own frequency shift in the 3He NMR signal. Arito thus replaced the copper demag stage with a Pomeranchuk refrigerator, which is based on the negative latent heat of solidification of 3He below 0.3 K. We then used the melting pressure itself as the thermomenter. While this should have given us a temperature resolution of 5-10 nK, instabilities in the 1K plate of the dilution refrigerator limited our temperature resolution to about 50 nK. However, this was entirely adequate for us to measure the frequency shift through the transition region. The shift itself is quite small, and one sees a constant maximum shift in the A1 phase just below Tc of only 3 Hz. To measure the frequency shift with sufficient precision, Arito use a feedback technique that allowed him to shift the magnitude of the static NMR field to stay on resonance. This required a special superconducting magnet in which one could couple flux into the main coil using a small superconducting DC transformer. I have owned such a magnet since 1974. The result of Arito's NMR frequency measurements show no frequency shift above Tc larger than about 0.1 Hz, with measurements having been carried out in magnetic fields from about 25 mT to over 400 mT. These are the only measurements in the A1 phase (where only the up-spins are paired) made below about 500 mT, and thus are of value beyond the obvious conclusion one can draw that critical fluctuations in liquid 3He near melting pressure (where they should be the largest) are consistent with the simple estimation that the critical regime should not be larger than about 1 nK.

	These results have been written up for publication, but we are waiting for Bruce Patton to add some of his calculations to our results prior to publication. Ironically, it was an artifact of bad measurement from the Wheatley group, which I was pursuing at Cornell, when I changed course in the middle of my experiment and discovered superfluidity in liquid 3He back in 1971. It does not appear that this investigation has been as fortunate. Arito Nisimori has now graduated and gone to a position with Quanum Design in Tokyo. The experiment he was to do for his Ph.D. thesis was the parity violation experiment, but this experiment had absolutely no need for microgravity, and thus was of less importance to this project. I had expected that my postdoc and graduate student on this project would share the cryostat, but that did not happen, and the postdoc maintained possession of the cryostat for the full three years of his stay here. None the less, I believe that this was a good experiment to try, and had the Wheatley results been correct, this would have provided a unique test of the renormalization group theory for critical phenomena.
Bibliography Type:	Description: (Last Updated: 02/27/2004)